

PIE Bashing

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Introduction

Many are introduced to this concept in the context of math competitions or a combo class. The general concept of a PIE-Bash can in fact be taken quite far, and it is quite fun. This note will employ a problem solving mindset to motivate a deeper exploration into this concept. Running into topics such as convolutions, atoms, posets, and lattices. Throughout the text there are many problems, all of them followed by solutions. The reader is encouraged to attempt to solve the problems, or at least come up with the idea, before reading the solution.

Simple PIE Bashing

We first consider the following problem.

Problem 2.1. Find the number of integers between 1 and 1000, inclusive, that are divisible by 3, 5, or 7.

Solution. To find the number of integers divisible by at least one of 3, 5, or 7, we first count the multiples of each individual number within the range from 1 to 1000:

$$\begin{aligned}\left\lfloor \frac{1000}{3} \right\rfloor &= 333, \\ \left\lfloor \frac{1000}{5} \right\rfloor &= 200, \\ \left\lfloor \frac{1000}{7} \right\rfloor &= 142.\end{aligned}$$

Summing these gives $333 + 200 + 142 = 675$. However, numbers that are divisible by more than one of these divisors have been counted multiple times. We correct for this by subtracting the numbers divisible by pairs of divisors:

$$\begin{aligned}\left\lfloor \frac{1000}{3 \times 5} \right\rfloor &= \left\lfloor \frac{1000}{15} \right\rfloor = 66, \\ \left\lfloor \frac{1000}{3 \times 7} \right\rfloor &= \left\lfloor \frac{1000}{21} \right\rfloor = 47, \\ \left\lfloor \frac{1000}{5 \times 7} \right\rfloor &= \left\lfloor \frac{1000}{35} \right\rfloor = 28.\end{aligned}$$

The sum of these pairwise intersections is $66 + 47 + 28 = 141$. Subtracting this from our running total gives $675 - 141 = 534$. Finally, numbers divisible by all three divisors (3, 5, and 7) were added three times and subtracted three times, meaning they have been completely excluded so far. We must add them back once:

$$\left\lfloor \frac{1000}{3 \times 5 \times 7} \right\rfloor = \left\lfloor \frac{1000}{105} \right\rfloor = 9.$$

Combining all these steps yields the final count:

$$675 - 141 + 9 = 543$$

□

The general idea is the following. Let

$$U_n := \{k \in \{1, 2, \dots, 1000\} \text{ which are divisible by } n\}$$

We need to compute

$$|U_3 \cup U_5 \cup U_7|$$

which from our solution turned out to be equal to:

$$|U_3 \cup U_5 \cup U_7| = (|U_3| + |U_5| + |U_7|) - (|U_3 \cap U_5| + |U_5 \cap U_7| + |U_3 \cap U_7|) + |U_3 \cap U_5 \cap U_7|$$

This motivates the following general theorem.

Theorem 2.2 (Principle of Inclusion-Exclusion). Let $A_1, A_2, \dots, A_n$ be finite sets. Then the size of their union is given by:

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n-1} |A_1 \cap A_2 \cap \dots \cap A_n|$$

which can be expressed compactly as:

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < i_2 < \dots < i_k \leq n} \left| \bigcap_{j=1}^k A_{i_j} \right|$$

Proof. We show that every element belonging to the union is counted exactly once. Consider an element x that is contained in k of the sets. Without loss of generality, let these sets be $A_1, A_2, \dots, A_k$. We proceed by induction on k , the number of sets containing x .

Base Case $k = 1$: If x is in only one set A_1 , it is counted once in the single-set sum $\sum |A_i|$ and does not appear in any pairwise or higher-order intersections. Thus, x is counted exactly 1 time.

Inductive Step: Assume that the claim holds for any element contained in $k - 1$ sets. Now, suppose x is contained in k sets: $A_1, A_2, \dots, A_k$.

We separate the collections of intersecting sets into two groups: those that do not contain A_k (formed from the first $k - 1$ sets) and those that contain A_k . For every collection of sets not containing A_k with size r , there is a corresponding collection containing A_k with size $r + 1$ formed by including A_k .

In the Principle of Inclusion-Exclusion, a collection of size r contributes with a sign of $(-1)^{r-1}$, while the corresponding collection of size $r + 1$ containing A_k contributes with a sign of $(-1)^{(r+1)-1} = (-1)^r$. Their combined contribution is:

$$(-1)^{r-1} + (-1)^r = 0$$

Thus, all such paired collections cancel each other out, with the single exception of the singleton set $\{A_k\}$ itself (which has size $r = 1$ and contributes $(-1)^{1-1} = 1$).

By the inductive hypothesis, the remaining collections account for the rest, and combined with this single additional contribution of 1 from $\{A_k\}$, the element x is counted a total of:

$$0 + 1 = 1$$

times. Therefore, every element in the union is counted exactly once.

□

An example application of this theorem is the following problem.

Problem 2.3. A postal worker has 6 distinct letters addressed to 6 different people, alongside n corresponding envelopes. If the worker places the letters into the envelopes completely at random (one letter per envelope), in how many ways can they do this such that **no** letter is placed into its correct envelope?

Solution. This is the classic problem of finding the number of *derangements*. The total number of ways to place n letters into n envelopes without any restrictions is simply $n!$. Let A_i be the set of arrangements where the i -th letter is placed into its correct envelope. We want to find the size of the union $|\bigcup_{i=1}^n A_i|$ to know how many invalid arrangements to subtract.

Consider the intersection of k of these sets, which represents fixing k specific letters in their correct envelopes. The remaining $n - k$ letters can be arranged in $(n - k)!$ ways. Since there are exactly $\binom{n}{k}$ ways to choose which k sets to intersect (i.e., which k letters are correctly placed), the sum of the sizes of all k -wise intersections is:

$$\sum_{i_1 < \dots < i_k} |A_{i_1} \cap \dots \cap A_{i_k}| = \binom{n}{k} (n - k)!$$

We can simplify this expression using the factorial definition of the binomial coefficient:

$$\binom{n}{k} (n - k)! = \frac{n!}{k!(n - k)!} (n - k)! = \frac{n!}{k!}$$

By the Principle of Inclusion-Exclusion, the number of arrangements with *at least one* correct letter is the alternating sum of these values from $k = 1$ to n :

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k-1} \frac{n!}{k!}$$

Finally, we subtract these invalid arrangements from the total number of unrestricted arrangements to find the number of ways where **no** letter is in the correct envelope:

$$n! - \sum_{k=1}^n (-1)^{k-1} \frac{n!}{k!} = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$$

Plugging in $n = 6$ gives 265 derangements. □

Advanced PIE Bashing

Suppose we consider the following problem.

Problem 3.1. Find the number of binary strings (using only 0s and 1s) of length 6 that are *aperiodic*—meaning they cannot be formed by repeating any strictly shorter substring multiple times.

Solution. Let A_d denote the set of length 6 binary strings formed by repeating a substring of length d . A string is periodic if it belongs to A_d for some proper divisor d of 6. The proper divisors of 6 are 1, 2, and 3. Notice that any string repeating every 1 character trivially repeats every 2 and 3 characters as well. Thus, $A_1 \subset A_2$ and $A_1 \subset A_3$, meaning the set of all periodic strings is exactly $A_2 \cup A_3$. We can apply the Principle of Inclusion-Exclusion to compute the total number of periodic strings:

$$|A_2 \cup A_3| = |A_2| + |A_3| - |A_2 \cap A_3|$$

To find the size of each set, we simply count the ways to freely choose the characters of the repeating block, which is 2^d . The intersection $A_2 \cap A_3$ represents strings that have both a period of 2 and a period of 3, which means they must have a period of $\gcd(2, 3) = 1$. Therefore:

$$\begin{aligned} |A_2| &= 2^2 = 4 \\ |A_3| &= 2^3 = 8 \\ |A_2 \cap A_3| &= |A_1| = 2^1 = 2 \end{aligned}$$

Substituting these values into our PIE formula yields the total number of periodic strings:

$$|A_2 \cup A_3| = 4 + 8 - 2 = 10$$

Subtracting these from the $2^6 = 64$ total unrestricted strings gives the number of aperiodic strings:

$$64 - 10 = 54$$

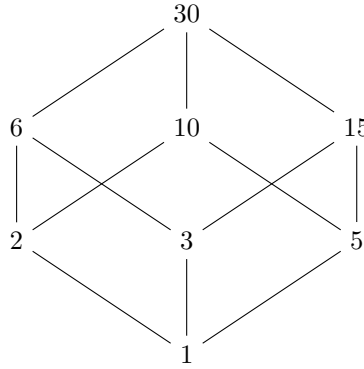
□

Well, that wasn't so hard, but what if I upgrade the problem. I dare you to solve the following with a standard PIE-Bash.

Problem 3.2. Find the number of binary strings (using only 0s and 1s) of length **30** that are *aperiodic*—meaning they cannot be formed by repeating any strictly shorter substring multiple times.

While the manual subtraction was easy, imagine trying to use the standard Principle of Inclusion-Exclusion for length 30. We might let A_d be the set of strings formed by repeating a block of length d . You would need to evaluate unions like $A_1 \cup A_2 \cup A_3 \cup A_5 \cup A_6 \cup A_{10} \cup A_{15}$.

In standard PIE, the intersection of sets represents elements belonging to all of them. Here, the intersection of strings built from length 2 blocks and length 3 blocks is the set of strings built from length $\gcd(2,3) = 1$ blocks. This means the intersections are better understood by the *poset of divisibility*, not subsets. The following is the poset of divisibility for $n = 30$.



We see that intersection of some A_i , is their *least lower bound*.

Standard PIE becomes a nightmare of redundant terms on a divisibility lattice. To elegantly handle inclusion-exclusion over divisors, we generalize the theorem into the *Möbius Inversion Formula*, which assigns a specific weight (the Möbius function, μ) to each divisor to cancel out overcounting more efficiently.

Definition 3.3 (Partially Ordered Set). A *partially ordered set* (or *poset*) consists of a set P along with a binary relation $\leq$ that formalizes the intuitive concept of an ordering. For the relation to be a partial order, it must satisfy the following three axioms for all $a, b, c \in P$:

Reflexivity: $a \leq a$

Antisymmetry: If $a \leq b$ and $b \leq a$, then $a = b$

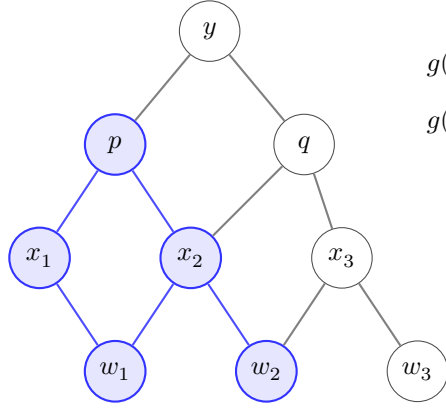
Transitivity: If $a \leq b$ and $b \leq c$, then $a \leq c$

It is "partial" because not every pair of elements needs to be comparable; there may exist elements x and y such that neither $x \leq y$ nor $y \leq x$ is true.

Suppose I have a poset P , and I have some function $f : P \rightarrow \mathbb{R}$ which I want to compute. Unfortunately, computing f might be too hard. On the other hand, suppose that I can easily compute

$$g(p) := \sum_{\substack{x \in P \\ x \leq p}} f(x)$$

for any $p \in P$. IE, I can easily compute the sums of lattice elements bounded above by a particular element:



$$g(p) = \sum_{x \leq p} f(x)$$

$$g(p) = f(p) + f(x_1) + f(x_2) + f(w_1) + f(w_2)$$

I want to find some function $\mu(x, y) : P \times P$ with the property that

$$f(p) = \sum_{\substack{x \in P \\ x \leq p}} \mu(x, p) g(x)$$

allowing me to compute f using my knowledge of g .

Problem 3.4. Suppose that P is the set of subsets of S , ordered by containment. Prove using the standard PIE Bash theorem that

$$\mu(x, y) = (-1)^{|y|-|x|}$$

satisfies the condition.

Lemma 3.5 (Möbius Inversion). For any locally finite poset P , there exists a unique function $\mu : P \times P \rightarrow \mathbb{R}$ such that for all $x \leq y$ in P ,

$$\sum_{x \leq z \leq y} \mu(z, y) = \delta_{x, y}$$

where $\delta_{x, y}$ is the Kronecker delta (equal to 1 if $x = y$ and 0 otherwise). If $x \not\leq y$, $\mu(x, y) = 0$.

Proof. We proceed constructively by induction on the size of the interval $[x, y] = \{z \in P \mid x \leq z \leq y\}$.

Base Case $x = y$: The condition requires $\sum_{x \leq z \leq x} \mu(z, x) = \delta_{x, x} = 1$. Since the only element in the interval $[x, x]$ is x itself, we must define:

$$\mu(x, x) = 1$$

Inductive Step ($x < y$): Assume $\mu(z, y)$ exists and has been uniquely determined for all z strictly between x and y . The condition for the interval $[x, y]$ requires:

$$\sum_{x \leq z \leq y} \mu(z, y) = \delta_{x, y} = 0$$

We can isolate the unknown term $\mu(x, y)$ by separating it from the rest of the sum:

$$\mu(x, y) + \sum_{x < z \leq y} \mu(z, y) = 0$$

Rearranging this equation provides a deterministic formula for $\mu(x, y)$:

$$\mu(x, y) = - \sum_{x < z \leq y} \mu(z, y)$$

Since P is locally finite, every interval contains a finite number of elements. Thus, this recursive construction is guaranteed to terminate. By construction, such a function μ satisfies the delta sum condition, proving both its existence and its uniqueness. $\square$

Theorem 3.6. For every function $f : P \rightarrow \mathbb{R}$ the function in Lemma 3.5 satisfies:

$$f(y) = \sum_{\substack{x \in P \\ x \leq y}} \mu(x, y) g(x) \quad g(y) := \sum_{\substack{x \in P \\ x \leq y}} f(x)$$

Proof. Consider some $p \leq y$. We need to consider the total contribution from $f(p)$ in the sum. Therefore we replace each occurrence of $g(x)$ with

$$I_p(x) := \begin{cases} f(p) & \text{if } p \leq x \\ 0 & \text{otherwise} \end{cases}$$

and the sum is

$$\begin{aligned} \sum_{\substack{x \in P \\ x \leq y}} \mu(x, y) I_p(x) &= \sum_{\substack{x \in P \\ p \leq x \leq y}} \mu(x, y) f(p) \\ &= f(p) \sum_{\substack{x \in P \\ p \leq x \leq y}} \mu(x, y) \\ &= f(p) \delta(p, y) \end{aligned} \quad \text{Lemma 3.5}$$

so the only contribution to the sum is $f(y)$ as desired. $\square$

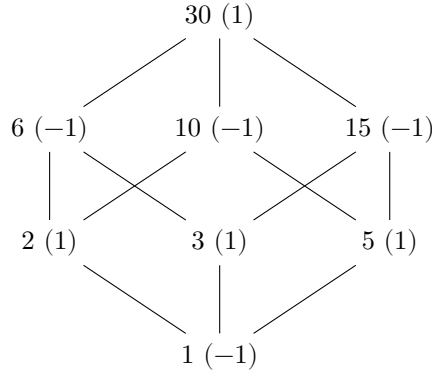
We now have enough machinery to return to the problem that was troubling us before.

Problem 3.7. Find the number of binary strings (using only 0s and 1s) of length 30 that are *aperiodic* (their fundamental period is exactly 30).

Solution. By the recurrence in the proof of Lemma 3.5

$$\mu(x, y) = - \sum_{x < z \leq y} \mu(z, y)$$

so value $\mu(d, 30)$ can be computed by taking the negative sum of the values strictly higher in the diagram:



As before, the number of ways to freely choose the characters of a repeating block is 2^d . By Möbius inversion, the total number of aperiodic strings is:

$$\sum_{d|30} \mu(d, 30) 2^d = 2^{30} - 2^{15} - 2^{10} - 2^6 + 2^5 + 2^3 + 2^2 - 2^1 = 1,073,708,010$$

$\square$

Despite the massive answer, the computation wasn't technically that hard. This illustrates the power of this formulism.

Incidence Algebras

This section is for those who enjoy finding the correct setting where a theorem has a very elegant statement/proof.

Definition 4.1 (Incidence Algebra). Let P be a locally finite poset. The *incidence algebra* $\mathcal{I}(P)$ is the set of all real-valued functions $f : P \times P \rightarrow \mathbb{R}$ such that $f(x, y) = 0$ whenever $x \not\leq y$.

To make this a true algebra, we equip it with a multiplication operation that mirrors our summation over intervals.

Definition 4.2 (Convolution). For any two functions $f, g \in \mathcal{I}(P)$, their *convolution* $f * g \in \mathcal{I}(P)$ is defined as:

$$(f * g)(x, y) = \sum_{x \leq z \leq y} f(x, z)g(z, y)$$

Theorem 4.3. [Associativity of the Incidence Algebra] Let P be a locally finite poset. For any functions $f, g, h \in \mathcal{I}(P)$, the convolution operation is associative, meaning $(f * g) * h = f * (g * h)$. Moreover the function

$$\delta(x, y) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases}$$

is the identity.

Proof. Let V be the free vector space spanned by the elements of P . We can reinterpret any function $f \in \mathcal{I}(P)$ as a linear transformation on this space by

$$f(p) = \sum_{x \in P} f(x, p)x \quad \text{for all } p \in P$$

which is well defined by the locally finite assumption.

Under this interpretation, convolution in the incidence algebra corresponds exactly to composition of functions. Because composition is associative, it immediately follows that convolution over the incidence algebra is also associative. Moreover δ corresponds exactly to the identity map $V \rightarrow V$. $\square$

Definition 4.4 (Zeta Function). The *zeta function* $\zeta \in \mathcal{I}(P)$ is defined as $\zeta(x, y) = 1$ if $x \leq y$ and 0 otherwise.

In our previous exposition we considered a function $f : P \rightarrow \mathbb{R}$ and wrote

$$g(p) := \sum_{\substack{x \in P \\ x \leq p}} f(x)$$

with the hope that g would be easier to compute. As in Theorem 4.3 we consider V as the free vector space spanned by P . Then we have induced maps

$$f, g : V \rightarrow \mathbb{R} \quad \zeta : V \rightarrow V$$

and the identification

$$g = f \circ \zeta$$

so that ζ is the function which constructs g from f .

Corollary 4.5 (Möbius Inversion as an Inverse). In the incidence algebra $\mathcal{I}(P)$, the zeta function ζ is

invertible, and its inverse is exactly the Möbius function μ . That is

$$\mu * \zeta = \delta = \zeta * \mu$$

so more explicitly:

$$\sum_{x \leq z \leq y} \mu(x, z) = \delta_{x, y} = \sum_{x \leq z \leq y} \mu(z, y)$$

Proof. Notice that the definition of the Möbius function in Lemma 3.5 is exactly the statement that:

$$\sum_{x \leq z \leq y} \mu(z, y) \zeta(x, z) = \delta(x, y)$$

In the language of the incidence algebra, this simply means $\zeta * \mu = \delta$. This gives us an abstraction of Möbius inversion. $\square$

Definition 4.6 (Ranked Poset). A poset P is *ranked* (or *graded*) if it can be equipped with a rank function $\rho : P \rightarrow \mathbb{N}$ satisfying two conditions:

- (a) If x is a minimal element of P , then $\rho(x) = 0$.
- (b) If y covers x (meaning $x < y$ and there exists no $z \in P$ such that $x < z < y$), then $\rho(y) = \rho(x) + 1$.

The rank of an element x is given by $\rho(x)$. If P has a maximum rank, this value is called the rank of the poset, denoted $\text{rank}(P)$.

Definition 4.7 (Characteristic Polynomial of a Poset). Let P be a finite ranked poset with a unique minimal element $\hat{0}$ and a rank function $\text{rank} : P \rightarrow \mathbb{N}$. The *characteristic polynomial* of P is defined as:

$$\chi(P, x) = \sum_{y \in P} \mu(\hat{0}, y) x^{\text{rank}(P) - \text{rank}(y)}$$

where μ is the Möbius function on the incidence algebra of P .

The characteristic polynomial serves as a counting theorem in various mathematical fields. Here are some notable applications:

Hyperplane Arrangements: When applied to the intersection poset of a set of hyperplanes, evaluations of the characteristic polynomial at specific values (such as $x = -1$ or $x = 1$) compute the exact number of total and bounded regions the hyperplanes divide the space into.

The Chromatic Polynomial: For a graph, evaluating this polynomial over the lattice of contractions (a poset formed by grouping vertices along edges) yields the Chromatic Polynomial. This determines the number of valid vertex colorings using k colors such that no adjacent vertices share the same color.

The Euler Characteristic: The evaluation of the Möbius function from the minimal element to the maximal element of a poset, denoted as $\mu(\hat{0}, \hat{1})$, represents the reduced Euler characteristic of the poset's order complex. This provides fundamental information about the topological "holes" in the corresponding geometric space.

Here is a problem which can be solved using this polynomial.

Problem 4.8 (2017 Putnam B6). Find the number of ordered 64-tuples $(x_0, x_1, \dots, x_{63})$ such that $x_0, x_1, \dots, x_{63}$ are distinct elements of $\{1, 2, \dots, 2017\}$ and

$$x_0 + x_1 + 2x_2 + 3x_3 + \dots + 63x_{63}$$

is divisible by 2017.

Solution. Let $p = 2017$, which is conveniently prime. We are looking for the number of injective functions $x : V \rightarrow \mathbb{F}_p$, where $V = \{0, 1, \dots, 63\}$, that satisfy a specific linear condition. Let $c_0 = 1$ and $c_i = i$ for $1 \leq i \leq 63$. Our sum condition is simply $\sum_{i \in V} c_i x_i \equiv 0 \pmod{p}$. A miraculous property of these coefficients is their total sum:

$$\sum_{i=0}^{63} c_i = 1 + \sum_{i=1}^{63} i = 1 + \frac{63 \times 64}{2} = 1 + 2016 = 2017 \equiv 0 \pmod{p}$$

Because the maximum possible coefficient is 63, the sum of any proper, non-empty subset of coefficients is strictly between 1 and 2016. Thus, for any subset $B \subseteq V$, $\sum_{i \in B} c_i \equiv 0 \pmod{p}$ if and only if $B = V$ or $B = \emptyset$.

Let Π_V be the poset of partitions of V , ordered by refinement. The minimal element $\hat{0}$ is the partition into 64 singletons, and the maximal element $\hat{1}$ is the single block V . Every function $f : V \rightarrow \mathbb{F}_p$ induces a partition $\ker(f) \in \Pi_V$ formed by the preimages of its values. The distinctness condition simply means we want to find the number of valid functions where $\ker(f) = \hat{0}$.

Let $f(\pi)$ be the number of valid functions with $\ker(f) = \pi$. Let $g(\pi) = \sum_{\sigma \geq \pi} f(\sigma)$ be the number of valid functions that are constant on the blocks of π .

To compute $g(\pi)$, we assign a single value $v_B \in \mathbb{F}_p$ to each block $B \in \pi$. The sum condition becomes:

$$\sum_{B \in \pi} \left(\sum_{i \in B} c_i \right) v_B \equiv 0 \pmod{p}$$

- **If $\pi = \hat{1}$:** There is only one block V , and its coefficient sum modulo p is 0. The equation reduces to $0 \cdot v_V \equiv 0 \pmod{p}$, which is satisfied by all p choices of v_V . Thus, $g(\hat{1}) = p$.
- **If $\pi \neq \hat{1}$:** At least one block has a non-zero coefficient sum modulo p . This forms a non-trivial linear equation over $\mathbb{F}_p$ with $|\pi|$ variables, which always has exactly $p^{|\pi|-1}$ solutions.

We can write this compactly for all $\pi \in \Pi_V$ as:

$$g(\pi) = p^{|\pi|-1} + (p-1)\delta_{\pi, \hat{1}}$$

By Möbius inversion on the incidence algebra of Π_V , we can extract $f(\hat{0})$:

$$\begin{aligned} f(\hat{0}) &= \sum_{\pi \in \Pi_V} \mu(\hat{0}, \pi) g(\pi) \\ &= \sum_{\pi \in \Pi_V} \mu(\hat{0}, \pi) (p^{|\pi|-1} + (p-1)\delta_{\pi, \hat{1}}) \\ &= \frac{1}{p} \sum_{\pi \in \Pi_V} \mu(\hat{0}, \pi) p^{|\pi|} + (p-1)\mu(\hat{0}, \hat{1}) \end{aligned}$$

We now prove two facts to evaluate this expression.

- Consider the number of injective functions $V \rightarrow \{1, 2, \dots, x\}$. This is clearly

$$x(x-1)(x-2) \dots (x-n+1)$$

by standard combinatorics. We could also compute this using a PIE-bash. To each partition π we associate the functions $V \rightarrow \{1, 2, \dots, x\}$ which are constant on the blocks of π . There are $x^{|\pi|}$ such functions. Therefore by Möbius inversion:

$$\sum_{\pi \in \Pi_n} \mu(\hat{0}, \pi) x^{|\pi|} = x(x-1)(x-2) \dots (x-n+1)$$

This is in fact x times the characteristic polynomial of the poset.

- By observing the x coefficient of the above polynomial we see that:

$$\mu(\hat{0}, \hat{1}) = (-1)^{n-1}(n-1)!$$

Substituting these identities into our inversion formula yields:

$$\begin{aligned} f(\hat{0}) &= \frac{1}{p} \left[p(p-1)(p-2) \dots (p-n+1) \right] + (p-1)(-1)^{n-1}(n-1)! \\ &= (p-1)(p-2) \dots (p-n+1) + (-1)^{n-1}(p-1)(n-1)! \end{aligned}$$

Finally, we plug in $p = 2017$ and $n = 64$:

$$f(\hat{0}) = \frac{2016!}{1953!} - 2016 \cdot 63!$$

□

This shows that sometimes, we do not need to explicitly calculate $\mu(x, y)$ for every $x \leq y$. We can instead reduce to a known problem.

Computing the Möbius Inversion

While this machinery is nice, in order to solve problems, it pays to have a better way to actually compute what μ is. Our current recursive bottom up method is simply unfeasible for large posets. Instead of an *algorithm*, what we need is a *formula*.

Theorem 5.1 (Philip Hall's Theorem). For any elements $x \leq y$ in a locally finite poset P , the Möbius function can be computed as:

$$\mu(x, y) = \sum_{k=0}^{\infty} (-1)^k c_k(x, y)$$

where $c_k(x, y)$ is the number of strict chains (or paths in the Hasse diagram) of length k between x and y . That is, the number of sequences $x = z_0 < z_1 < \dots < z_k = y$.

Proof. Let $F(x, y) = \sum_{k=0}^{\infty} (-1)^k c_k(x, y)$. We will show that $F(x, y)$ satisfies the recursive definition of $\mu(x, y)$ established in Lemma 3.5.

Base Case ($x = y$): The only valid chain from x to x is the trivial chain $x = z_0$ of length $k = 0$. Thus, $c_0(x, x) = 1$, and $c_k(x, x) = 0$ for all $k \geq 1$. This gives:

$$F(x, x) = (-1)^0(1) = 1$$

which matches our required base case $\mu(x, x) = 1$.

Inductive Step ($x < y$): For $x < y$, there are no chains of length 0, meaning $c_0(x, y) = 0$. For any $k \geq 1$, every chain of length k from x to y must pass through some penultimate element z strictly before y , where $x \leq z < y$. Thus, we can recursively decompose the number of chains:

$$c_k(x, y) = \sum_{x \leq z < y} c_{k-1}(x, z)$$

Substituting this identity into our formula for $F(x, y)$ yields:

$$\begin{aligned}
F(x, y) &= \sum_{k=1}^{\infty} (-1)^k \left(\sum_{x \leq z < y} c_{k-1}(x, z) \right) \\
&= - \sum_{x \leq z < y} \sum_{k=1}^{\infty} (-1)^{k-1} c_{k-1}(x, z) \\
&= - \sum_{x \leq z < y} \sum_{j=0}^{\infty} (-1)^j c_j(x, z) \\
&= - \sum_{x \leq z < y} F(x, z)
\end{aligned}$$

Notice that $F(x, y)$ satisfies the exact same base case and recurrence relation as $\mu(x, y)$. Since Lemma 3.5 established that this recursive construction defines a strictly unique function, it must be true that $F(x, y) = \mu(x, y)$. $\square$

Problem 5.2. Let $V = \{0, 1\}^4$, where addition is performed coordinatewise modulo 2. A subset

$$A \subseteq V \setminus \{0\}$$

is called *complete* if every element of V can be expressed as a sum of distinct elements of A . How many complete subsets of $V \setminus \{0\}$ are there?

Solution. We regard V as a four-dimensional vector space over $\mathbb{F}_2$. Since every coefficient in $\mathbb{F}_2$ is either 0 or 1, the condition that every vector is a sum of distinct elements of A is equivalent to

$$\text{span}(A) = V.$$

Let P be the poset of linear subspaces of V , ordered by containment. For every subspace $U \leq V$, define:

$$f(U) := \# \{A \subseteq U \setminus \{0\} \mid \text{span}(A) = U\} \quad g(U) := \# \{A \subseteq U \setminus \{0\}\}.$$

Every subset of $U \setminus \{0\}$ has a unique span $W \leq U$, so

$$g(U) = \sum_{W \leq U} f(W).$$

If $\dim U = d$, then U has $2^d - 1$ nonzero elements, and hence

$$g(U) = 2^{2^d - 1}.$$

Möbius inversion on the subspace poset therefore gives

$$f(V) = \sum_{U \leq V} \mu(U, V) 2^{2^{\dim U} - 1}.$$

It remains to compute the necessary values of $\mu(U, V)$. The interval $[U, V]$ is isomorphic to the lattice of subspaces of the quotient space V/U . Consequently, $\mu(U, V)$ depends only on

$$r = \dim(V/U) = 4 - \dim U.$$

Write this common value as m_r . By Philip Hall's theorem,

$$m_r = \sum_{k=0}^r (-1)^k c_k(r),$$

where $c_k(r)$ is the number of strict chains

$$\{0\} = W_0 < W_1 < \dots < W_k = \mathbb{F}_2^r.$$

We now count these chains. From a known combinatorial result over $\mathbb{F}_2$ (or, you can compute this yourself):

- A two-dimensional vector space over $\mathbb{F}_2$ has 3 lines.
- A three-dimensional vector space has 7 lines and 7 planes.
- A four-dimensional vector space has 15 lines, 35 planes, and 15 hyperplanes.

It follows that

r	$c_0(r)$	$c_1(r)$	$c_2(r)$	$c_3(r)$	$c_4(r)$	m_r
0	1	0	0	0	0	1
1	0	1	0	0	0	-1
2	0	1	3	0	0	2
3	0	1	14	21	0	-8
4	0	1	65	315	315	64

Indeed, in dimension 4 we have

$$c_2(4) = 15 + 35 + 15 = 65.$$

A chain with two intermediate subspaces has dimension sequence

$$(1, 2), \quad (1, 3), \quad \text{or} \quad (2, 3).$$

There are respectively

$$35 \cdot 3, \quad 15 \cdot 7, \quad 15 \cdot 7$$

such chains, so

$$c_3(4) = 105 + 105 + 105 = 315.$$

Finally, the number of complete flags

$$\{0\} < L < P < H < V$$

is

$$15 \cdot 7 \cdot 3 = 315,$$

giving $c_4(4) = 315$. The numbers of subspaces of V of dimensions 0, 1, 2, 3, 4 are

$$1, 15, 35, 15, 1,$$

respectively. Thus

$$\begin{aligned} f(V) &= 1 \cdot 64 \cdot 1 + 15 \cdot (-8) \cdot 2 + 35 \cdot 2 \cdot 8 + 15 \cdot (-1) \cdot 128 + 1 \cdot 1 \cdot 32768 \\ &= 64 - 240 + 560 - 1920 + 32768 \\ &= 31232 \end{aligned}$$

is the number of complete subsets. This is quite a lot. □

It turns out that the Mobius function for the poset of subspaces of an n dimensional $\mathbb{F}_p$ -vector space is:

$$\mu(U, V) = (-1)^k p^{\binom{k}{2}} \quad k := \dim V - \dim U$$

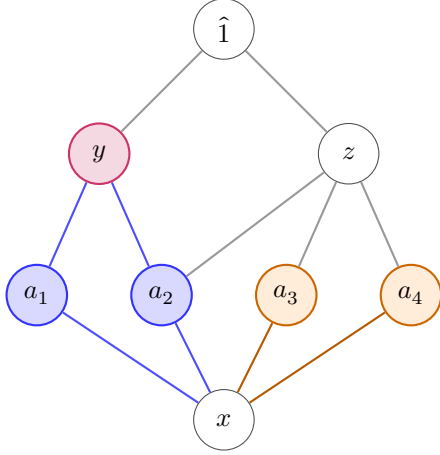
Unfortunately this is quite hard to prove. We can learn to approach it more effectively with some more machinery.

Definition 5.3 (Lattice). A *lattice* is a specific type of poset in which every pair of elements $x, y \in P$ has a unique least upper bound and a unique greatest lower bound. Specifically:

- (a) The *join* (or supremum, least upper bound), denoted $x \vee y$, exists in P .
- (b) The *meet* (or infimum, greatest lower bound), denoted $x \wedge y$, exists in P .

For example, consider the poset of integers ordered by divisibility (where $a \leq b$ means a divides b). The structure forms a lattice where the join $a \vee b$ is the least common multiple ($\text{lcm}(a, b)$) and the meet $a \wedge b$ is the greatest common divisor ($\text{gcd}(a, b)$). It turns out that when our poset is a lattice, there is another slick formula for μ which is sometimes easier to compute.

Definition 5.4 (Atoms). In a poset with a bottom element x , an *atom* relative to x is an element a that directly covers x . That is, $x < a$ and there is no element z such that $x < z < a$. Let $\mathcal{A}_x$ denote the set of all atoms covering x .



$$\mathcal{A}_x = \{a_1, a_2, a_3, a_4\} \quad (\text{Atoms covering } x)$$

$$S = \{a_1, a_2\} \subseteq \mathcal{A}_x \quad (\text{Selected subset of atoms})$$

$$\bigvee S = a_1 \vee a_2 = y \quad (\text{Join of the subset } S)$$

Theorem 5.5 (Rota's Crosscut Theorem). Let P be a finite lattice, and let $x, y \in P$ with $x \leq y$. If $\mathcal{A}_x$ is the set of atoms covering x , then the Möbius function is given by:

$$\mu(x, y) = \sum_{\substack{S \subseteq \mathcal{A}_x \\ \bigvee S = y}} (-1)^{|S|}$$

where $\bigvee S$ denotes the join of the elements in S . By convention, the join of the empty set is x .

Proof. We define a function $f(y)$ on the interval $[x, y]$ corresponding to the Crosscut sum:

$$f(y) = \sum_{\substack{S \subseteq \mathcal{A}_x \\ \bigvee S = y}} (-1)^{|S|}$$

Recall from Lemma 3.5 that the Möbius function is uniquely determined by the property that for all $x \leq y$:

$$\sum_{x \leq z \leq y} \mu(x, z) = \delta_{x, y}$$

We will show that $f(y)$ satisfies this exact same recurrence. Summing $f(z)$ over all z in the interval $[x, y]$, we have:

$$\sum_{x \leq z \leq y} f(z) = \sum_{x \leq z \leq y} \sum_{\substack{S \subseteq \mathcal{A}_x \\ \bigvee S = z}} (-1)^{|S|}$$

Notice that this double sum partitions all subsets $S \subseteq \mathcal{A}_x$ by their resulting join z . Summing over all $z \in [x, y]$ is therefore mathematically identical to summing over all subsets S whose join is less than or equal to y . Thus, we can condense the sum to:

$$\sum_{x \leq z \leq y} f(z) = \sum_{\substack{S \subseteq \mathcal{A}_x \\ \bigvee S \leq y}} (-1)^{|S|}$$

Let $\mathcal{A}_{x, y} = \{a \in \mathcal{A}_x \mid a \leq y\}$ be the restricted set of atoms covering x that are bounded above by y . For any subset $S \subseteq \mathcal{A}_x$, the condition $\bigvee S \leq y$ is satisfied if and only if every individual atom in S is less than or equal to y . In other words, S must be a subset of $\mathcal{A}_{x, y}$. This allows us to re-index the sum purely over a power set:

$$\sum_{x \leq z \leq y} f(z) = \sum_{S \subseteq \mathcal{A}_{x, y}} (-1)^{|S|}$$

This is exactly a PIE-bash! By the binomial theorem, this alternating sum over the power set of $\mathcal{A}_{x,y}$ evaluates directly to $(1-1)^{|\mathcal{A}_{x,y}|}$. If $x = y$, there are no elements strictly between x and y , so $\mathcal{A}_{x,x} = \emptyset$. The sum over the empty set gives $(-1)^0 = 1$. If $x < y$, there exists at least one atom below y (since P is a finite lattice), making $|\mathcal{A}_{x,y}| \geq 1$, which forces the sum to evaluate to 0. Therefore, we arrive at:

$$\sum_{x \leq z \leq y} f(z) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x < y \end{cases} = \delta_{x,y}$$

Because $f(y)$ satisfies the identical sum condition as the Möbius function, and Lemma 3.5 establishes that this recurrence defines a strictly unique function, we conclude that $f(y) = \mu(x, y)$. $\square$

We can now attack the following problem. Despite looking entirely impossible from a naive point of view, the idea of the crosscut makes it very approachable.

Problem 5.6. Let G be a finite group whose Frattini subgroup $\Phi(G)$ (defined as the intersection of all maximal subgroups) is non-trivial ($\Phi(G) \neq \{1\}$). Consider all strictly increasing chains of subgroups from the trivial group to the entire group:

$$\{1\} = H_0 < H_1 < H_2 < \dots < H_k = G$$

Let $\mathcal{C}_{\text{even}}$ be the number of such chains of even length k , and let $\mathcal{C}_{\text{odd}}$ be the number of such chains of odd length k . Prove that $\mathcal{C}_{\text{even}} = \mathcal{C}_{\text{odd}}$.

Solution. By Theorem 5.1 we note:

$$\mathcal{C}_{\text{even}} - \mathcal{C}_{\text{odd}} = \mu(\{1\}, G)$$

To evaluate this using Rota's Crosscut Theorem, we look at the upper crosscut formed by the set $\mathcal{M}$ of all maximal subgroups of G . Since every proper subgroup is contained in at least one maximal subgroup, $\mathcal{M}$ forms a valid upper crosscut.

By the dual version of Rota's Crosscut Theorem, the Möbius function is given by an alternating sum over subsets of the crosscut whose intersection equals the bottom element:

$$\mu(\{1\}, G) = \sum_{\substack{S \subseteq \mathcal{M} \\ \bigcap_{M \in S} M = \{1\}}} (-1)^{|S|}$$

By definition, the Frattini subgroup $\Phi(G)$ is the intersection of all maximal subgroups in $\mathcal{M}$. Because we are given that $\Phi(G) \neq \{1\}$, the intersection of any collection of maximal subgroups must contain $\Phi(G)$ and therefore cannot equal $\{1\}$. Thus, no subset $S \subseteq \mathcal{M}$ satisfies the intersection condition, making the sum empty:

$$\mu(\{1\}, G) = 0 \implies \mathcal{C}_{\text{even}} = \mathcal{C}_{\text{odd}}$$

$\square$

Theorem 5.7 (Weisner). Let L be a finite lattice with minimum $\hat{0}$ and maximum $\hat{1}$. For any $a \in L \setminus \{\hat{0}\}$:

$$\sum_{\substack{x \in L \\ x \vee a = \hat{1}}} \mu(\hat{0}, x) = 0$$

Proof. In the incidence algebra $\mathcal{I}(L)$ define the diagonal projection $D_a \in \mathcal{I}(L)$ by:

$$D_a(x, y) := \delta_{x,y} \mathbb{1}_{x \geq a}$$

Since $z \geq x$ and $z \geq a$ if and only if $z \geq x \vee a$, left-multiplying by ζ yields $(\zeta * D_a)(x, z) = \zeta(x \vee a, z)$. Right-multiplying by μ constructs the join-indicator element $W_a := \zeta * D_a * \mu$, whose entries satisfy

$$W_a(x, y) = \sum_{x \leq z \leq y} \zeta(x \vee a, z) \mu(z, y) = (\zeta * \mu)(x \vee a, y) = \delta(x \vee a, y)$$

Therefore

$$\mu * W_a(\hat{0}, \hat{1}) = \sum_{\hat{0} \leq x \leq \hat{1}} \mu(\hat{0}, x) \delta(x \vee a, y) = \sum_{x \vee a = \hat{1}} \mu(\hat{0}, x) = 0$$

so sum in question is precisely the $(\hat{0}, \hat{1})$ entry of $\mu * W_a$. By associativity and the identity $\mu * \zeta = \delta$, the expression dissolves:

$$\sum_{x \vee a = \hat{1}} \mu(\hat{0}, x) = (\mu * W_a)(\hat{0}, \hat{1}) = (\mu * \zeta * D_a * \mu)(\hat{0}, \hat{1}) = (D_a * \mu)(\hat{0}, \hat{1}) = D_a(\hat{0}, \hat{0})\mu(\hat{0}, \hat{1}).$$

Because $a > \hat{0}$, we have $\hat{0} \not\leq a$, whence $D_a(\hat{0}, \hat{0}) = 0$ and the sum is 0. $\square$

Now we can solve the following problem. This would have simplified our solution to Problem 5.2.

Proof. Prove that the Mobius function for the poset of subspaces of an $\mathbb{F}_p$ -vector space is:

$$\mu(U, V) = (-1)^k p^{\binom{k}{2}} \quad k := \dim V - \dim U$$

$\square$

Proof. Working in the quotient V/U , we see that $\mu(U, V)$ only depends on $\dim V - \dim U$. Therefore we desire to compute:

$$\mu(\{0\}, V) \quad \dim V = k$$

We will use induction on k .

Base Case $k = 1$: In this case we can use Philip Hall's theorem. There is only a single path of length 1 from $\{0\}$ to V so $\mu(\{0\}, V) = -1$ which agrees with the formula $(-1)^k p^{\binom{k}{2}}$.

Inductive Step: Assume the formula holds for all $\mathbb{F}_p$ -vector spaces of dimension $k-1$ where $k \geq 2$. Fix some nonzero one-dimensional linear subspace $l \subseteq V$. Note that this poset has unique largest and smallest elements $\{0\}$ and V . Moreover the join is the sum of subspaces, and the meet is the intersection. Therefore the conditions of Weisner's theorem are satisfied and

$$\sum_{\substack{\text{subspace } U \subseteq V \\ U+l=V}} \mu(\{0\}, U) = 0 \quad \implies \quad \mu(\{0\}, V) = - \sum_{\substack{\text{proper subspace } U \subsetneq V \\ U+l=V}} \mu(\{0\}, U)$$

It is clear any such U must be a hypersurface of codimension 1 in V . By a standard result there are

$$\binom{k}{k-1}_p = \frac{p^k - 1}{p - 1}$$

such hypersurfaces. We only want the ones which do not contain l (so that $U+l=V$). The ones which do contain l are identified with hypersurfaces in V/l of which there are $\frac{p^{k-1}-1}{p-1}$. Therefore there are

$$\frac{p^k - 1}{p - 1} - \frac{p^{k-1} - 1}{p - 1} = p^{k-1}$$

relevant hypersurfaces. Therefore by induction

$$\begin{aligned} \mu(\{0\}, V) &= - \sum_{\substack{\text{proper subspace } U \subsetneq V \\ U+l=V}} \mu(\{0\}, U) \\ &= -p^{k-1}(-1)^{k-1} p^{\binom{k-1}{2}} \\ &= (-1)^k p^{\binom{k}{2}} \end{aligned}$$

as desired. $\square$

Congrats, you now know an unusual amount about how to PIE-bash correctly.